

Directory Pipeline

(A personal digital collections project)

2026-09-13

All Eyes on AI for LAMs: IIF & IIPC

Josh Hadro

josh.hadro@gmail.com



New York
Public
Library



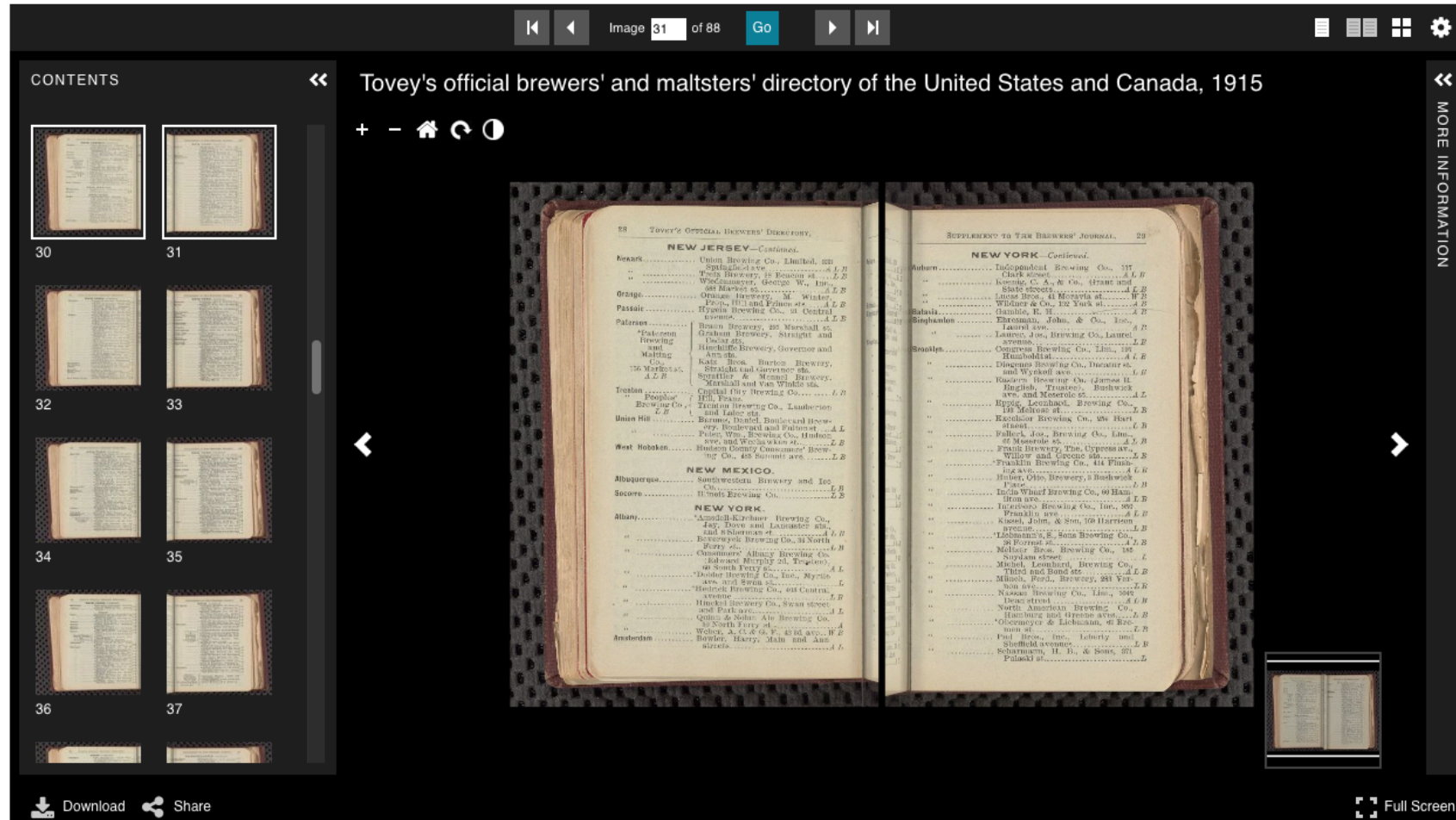
LIBRARY **LIBRARY
OF CONGRESS**

What the pipeline does

Takes any item with [a IIIF manifest](#) and runs through a human-in-the-loop, Large Language Model (LLM)-augmented set of steps with meta-prompting to produce structured data with minimal customization required.

The primary output: An HTML/JS table viewer (the Data Explorer) with one row per entry in the source material, including images of the location of the entry on the page and a link back to the originating digital object.

Tovey's official brewers' and maltsters' directory of the United States and Canada, 1915



Our collections include some content that may be harmful or difficult to view. [Learn more.](#)

Data source:

[Research Catalog](#)

Purchase this print

[Order Print](#)

Image ID

5131484

PIPELINE DATA EXPLORER

Tovey's official brewers' and maltsters' directory of the United States and Canada, 1915

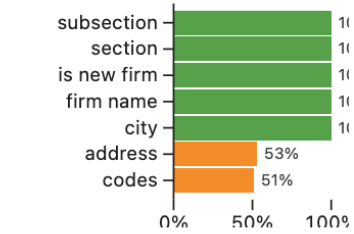
1,642 entries

SELECT VOLUME

Tovey's official brewers' and ▾

FIELD COVERAGE

Less than 100% means not every entry from the source scans had that field or value.



PRODUCT CODES

- ☐ LB11
- ☐ ALB8
- ☐ L B0
- ☐ L4
- ☐ AL1
- ☐ A1
- ☐ AB0
- ☐ A L B0
- ☐ WB1
- ☐ ALWB0

+ 15 more...

IS MALT HOUSE

- ☐ No23
- ☐ Yes-

Search all fields (name, address, city...)

26 of 1,642 entries

✕ Clear filters

1915

The New York Public Library

Directories

Item page ↗

ⓘ About

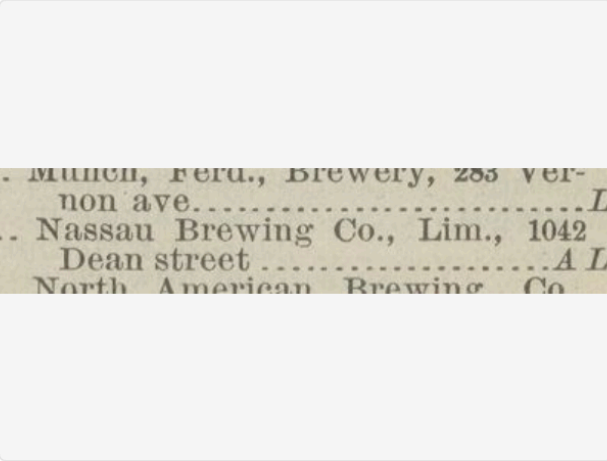
Show summary ▾

Export CSV

state ▴	city ▴	business name ▴	address ▴	product codes ⓘ ▴	is malt house ⓘ ▴
NEW YORK	Brooklyn	Excelsior Brewing Co.	237 Hart street	LB	No
NEW YORK	Brooklyn	Fallett, Jos., Brewing Co., Lim.	66 Meserole st	ALB	No
NEW YORK	Brooklyn	Frank Brewery, The	Cypress av., Willow and Greene sts.	LB	No
NEW YORK	Brooklyn	Franklin Brewing Co.	414 Flushing ave.	ALB	Yes
NEW YORK	Brooklyn	Huber, Otto, Brewery	3 Bushwick Place	LB	No
NEW YORK	Brooklyn	India Wharf Brewing Co.	60 Hamilton ave.	ALB	No
NEW YORK	Brooklyn	Interboro Brewing Co., Inc.	950 Franklin ave	ALB	No
NEW YORK	Brooklyn	Kissel, John, & Son	169 Harrison avenue	LB	No
NEW YORK	Brooklyn	Liebmann's, S., Sons Brewing Co.	36 Forrest st.	ALB	Yes
NEW YORK	Brooklyn	Meltzer Bros. Brewing Co.	185 Suydam street	L	No
NEW YORK	Brooklyn	Michel, Leonhard, Brewing Co.	Third and Bond sts	ALB	No
NEW YORK	Brooklyn	Münch, Ferd., Brewery	283 Vernon ave	L	No
NEW YORK	Brooklyn	Nassau Brewing Co., Lim.	1042 Dean street	ALB	No
NEW YORK	Brooklyn	North American Brewing Co.	Hamburg and Greene aves	LB	No
NEW YORK	Brooklyn	Obermeyer & Liebmann	67 Bremen st.	LB	Yes
NEW YORK	Brooklyn	Piel Bros., Inc.	Liberty and Sheffield avenues	LB	No
NEW YORK	Brooklyn	Scharmann, H. B., & Sons	371 Pulaski st	L	No
NEW YORK	Brooklyn	Strasser & Schwing	174 North 9th street	WB	No
NEW YORK	Brooklyn	Streeter & Denison	Wythe and Metropolitan aves	A	No
NEW YORK	Brooklyn	Trommer, John F., Evergreen Brewery	1632 Bushwick ave	LB	No
NEW YORK	Brooklyn	Ulmer, Wm., Brewery, Inc.	29 Belvidere st	L	No
NEW YORK	Brooklyn	Welz & Zerweck, Inc.	Myrtle and Wyckoff aves.	LB	No

Showing all 26 entries

Nassau Brewing Co., Lim. ✕



state	NEW YORK
city	Brooklyn
business name	Nassau Brewing Co., Lim.
address	1042 Dean street
product codes	ALB
is malt house	No
is new listing	No

[NYPL Digital Collections ↗](#) · [View in source document ↗ \(exact line\)](#)

Top takeaways — what's new here for digital collections?

- Meta-prompting strategy for custom, item-specific OCR and NER prompts based on sample pages
- Two-pass consensus OCR strategy: old tech for bounding boxes, LLMs for higher-quality OCR, matched via alignment step
 - A critical guard against hallucinations

The pipeline

Download files

- Select sample pages*
- Generate OCR + NER prompts
- Run LLM OCR
- Run layout detection
- Match lines
- Review alignment*
- Extract aligned entry data

* denotes "human in the loop" step, through Pipeline interface

Then the fun stuff

- Data explorer
- Map interface (after geocoding enrichment)
- Cross-volume comparison
- And more!

Example outputs

- [The Green Books + Other Travel Guide Explorer \(1930-1966\)](#)
 - N.B. Lots of design elements layered onto this data explorer
- [Date Explorer for 19th Century brewery guides, extracted from a brewery guide](#)
 - Closer to raw wireframe output
- [An exploration tool for data from the Polk-Hoffhine Tulsa City Directory, 1921](#)

A DIRECTORY OF PLACES THAT WELCOMED BLACK TRAVELERS, 1930 – 1966.

The Negro Motorist Green Book wasn't the only travel guide for African Americans listing hotels, restaurants, beauty parlors, and other businesses safe to visit during the Jim Crow era of segregation. This explorer makes records searchable across every digitized volume of eight different publications.

START EXPLORING →

OPEN A RANDOM ENTRY

Source material held by the New York Public Library, Schomburg Center for Research in Black Culture, and the Library of Congress. Data extracted using an LLM-enhanced [Directory Pipeline](#) for OCR and named-entity recognition.

A NOTE ON ACCURACY

This dataset carries occasional OCR, entity-extraction, and entry-alignment errors, so always treat the original published pages as the authoritative source. This explorer is a way to engage with the directories more deeply, *not a replacement for them*. Data has not been reviewed for 100% accuracy.

[READ MORE ABOUT THE DATA AND EXPLORER→](#)

Start exploring with a random entry:

SHUFFLE



Frank's Restaurant

1959 · NEW YORK, NEW YORK

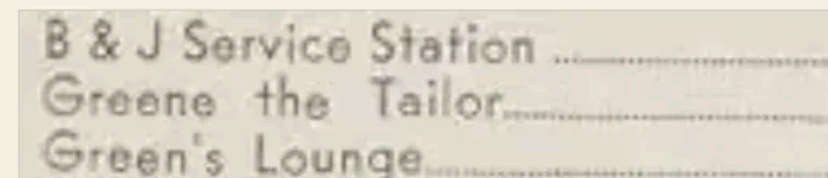
RESTAURANTS



COLDEN'S RESTAURANT

1956 · MIAMI, FLORIDA

RESTAURANTS



Greene the Tailor

1966 · PEEKSKILL, NEW YORK

TAILOR SHOPS

Every entry links to its exact location on the page.

GREEN BOOKS + OTHER TRAVEL GUIDE EXPLORER

Explore the listings of safe businesses and services for Black travelers, 1930–1966

ABOUT THIS EXPLORER

CHARTS & TRENDS

[OPEN A RANDOM ENTRY](#)

Dew Drop

LISTINGS
ESTABLISHMENTS
101 of 113,827 entries
x CLEAR FILTERS

SEARCH: Dew Drop x

YEAR :	PUBLICATION :	NAME :	ADDRESS :
↓ ↑ ↻	(all)	filter...	filter...
1938	The Negro Motorist Green Book	Dew Drop Inn	E. Locust St.
1938	The Negro Motorist Green Book	Dew Drop	363 Halsey St.
1939	The Negro Motorist Green Book	Dew Drop Ir	11 Ashby N. W.
1939	The Negro Motorist Green Book	Dew Drop Inn	E. Locust St.
1939	The Negro Motorist Green Book	Dew Drop	363 Halsey St.
1940	The Negro Motorist Green Book	Dew Drop Inn	11 Ashby St. N. W
1940	The Negro Motorist Green Book	Dew Drop Inn	E. Locust St.
1940	The Negro Motorist Green Book	Dew Drop	363 Halsey St.
1941	The Negro Motorist Green Book	Dew Drop Inn	11 Ashby St. N. W
1941	The Negro Motorist Green Book	Dew Drop Inn	E. Locust St.
1941	The Negro Motorist Green Book	Dew Drop	363 Halsey St.
1941	The Negro Motorist Green Book	Dew Drop Inn	2514 Booker St.

FILTER BY

Year

- ☐ 1930 1
- ☐ 1931 1
- ☐ 1937 0
- ☐ 1938 2
- ☐ 1939 3
- ☐ 1940 4
- ☐ 1941 4
- ☐ 1946 5
- ☐ 1947 10
- ☐ 1948 6
- + 16 more...

Publication

- ☐ ● The Negro Motorist G... 81
- ☐ ● Travelguide 7
- ☐ ● Go, Guide to Pleasant ... 9

[View in page context ↗](#)

Dew Drop Inn

<
>
15 of 101
COPY LINK

WHERE

ADDRESS 2836 La Salle St.

LOCATION NEW ORLEANS, LOUISIANA

AS PRINTED

CATEGORY NIGHT CLUBS

● The Negro Motorist Green Book · 1946

COVERAGE ACROSS PUBLICATIONS

Filed under 5 section headings across guides: NIGHT CLUBS, MISCELLANEOUS, NIGHT CLUB - RESTAURANTS, HOTELS, ...

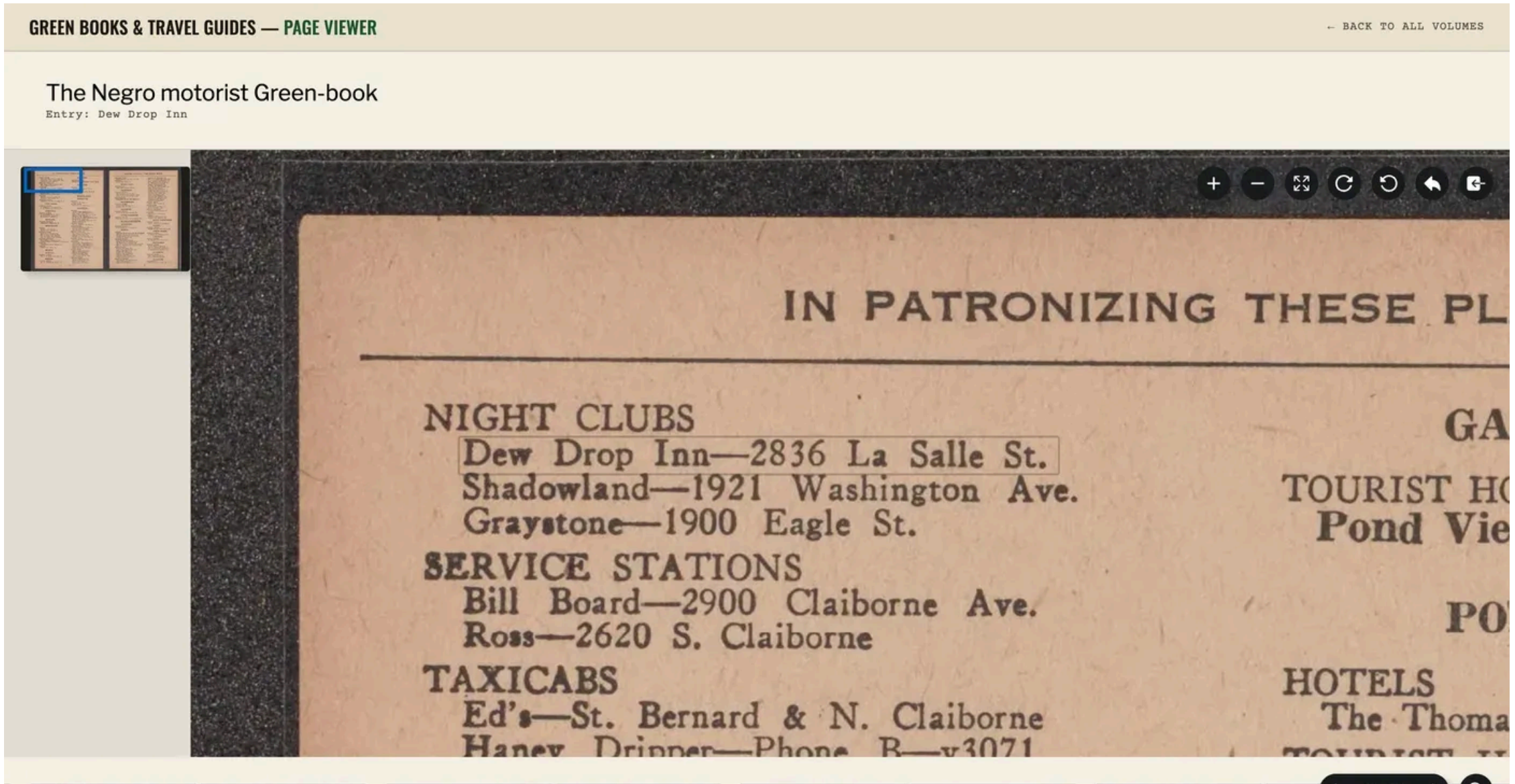
18 listings across **3 of 8** publications · **1946–1955**

- Hackley's
- Travelers Guide
- Green Book
- Smith's
- Travelguide
- Go Guide
- Afro-American
- NHA Directory

1942–1945: no guides in the dataset were published — the wartime pause.

■ Listed — colored by publication □ Published, not listed

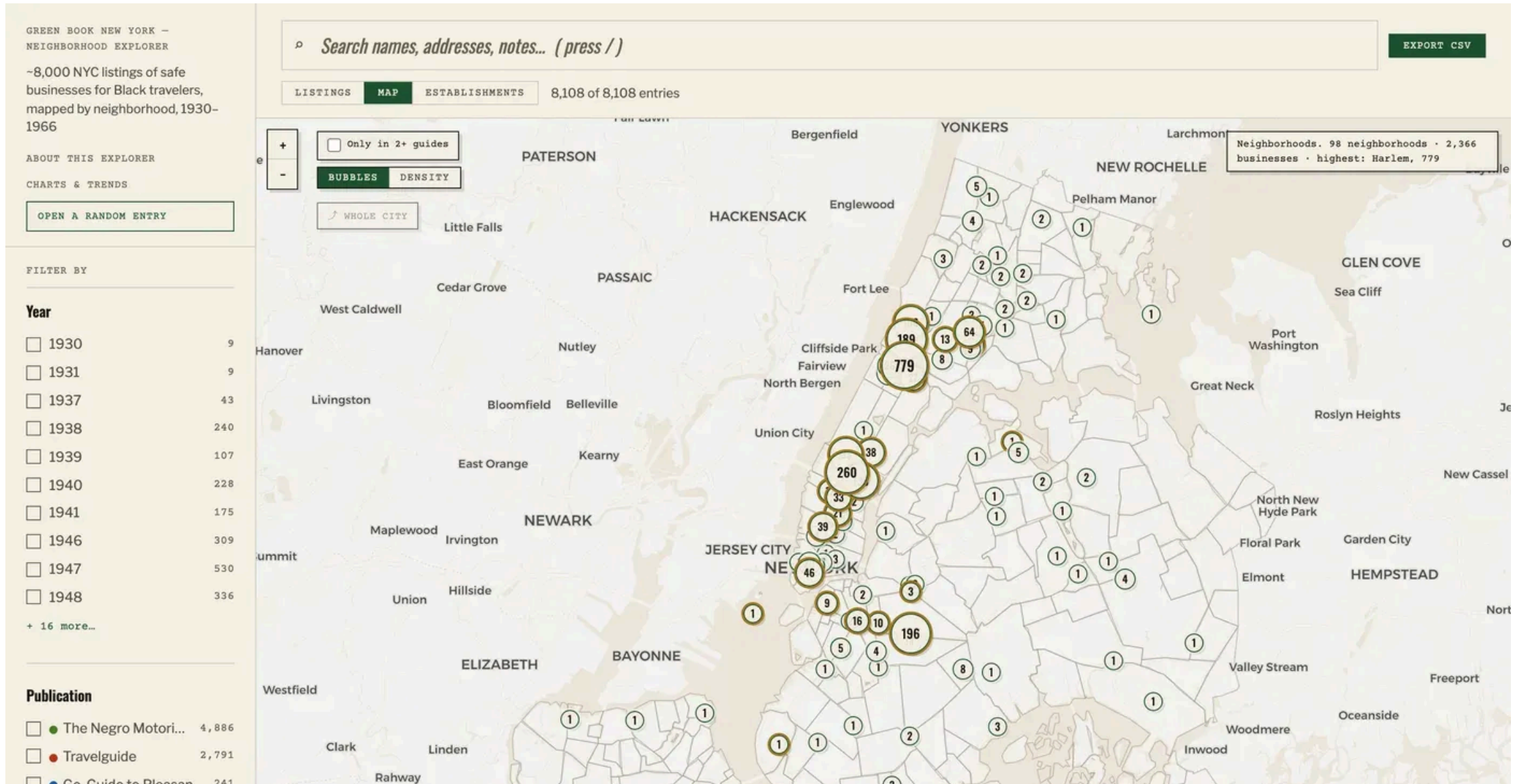
IIIF viewer that zooms directly to the entry in context



Cross-volume views

[illegible]

A New York City map version



Appendices

Human-in-the-loop interface examples

- Export entry boxes

○ Build ranges

(pending)

— https://www.loc.gov/item/73644404/

output/woods_directory_73644404 model: gemini-2.0-flash

×

INGEST

✓ Download

CALIBRATE

○ Select pages ↗

✓ Generate prompts

Sends selected sample pages to Gemini and generates volume-specific ocr_prompt.md and ner_prompt.md. Run once per collection type; reuse prompts for subsequent volumes.

Model: gemini-2.0-flash

Gemini model used to analyze sample pages and write the prompts.

OCR only: ☐ Generate only ocr_prompt.md; skip the NER prompt.

NER only: ☐ Generate only ner_prompt.md; skip the OCR prompt.

python main.py https://www.loc.gov/item/73644404/ --generate-prompts --model gemini-2.0-flash

Force-run

cancel

○ Generate palette

OCR

✓ Surya OCR

✓ Gemini OCR

✓ Align OCR

REVIEW

○ Review alignment ↗

EXTRACT

✓ Extract entries

✓ Explore

✓ Geocode

✓ Map

IIIF

○ Export annotations

✓ Export entry boxes

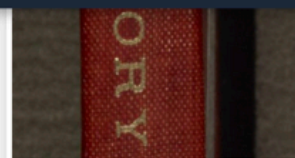
○ Build ranges

(pending)

— https://archive.org/details/johleonhardfrisc01fris_0/page/36/mode/2up

output/joh_leonhard_frisch_beschreibung_von_all_johleonhardfrisc01fr model: gemini-2.0-flash

×



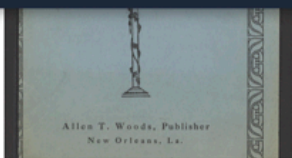
p. 1



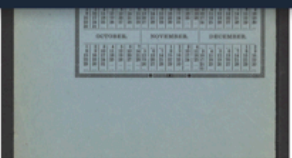
p. 2



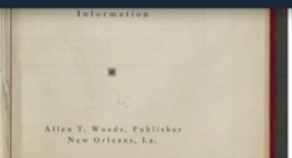
p. 3



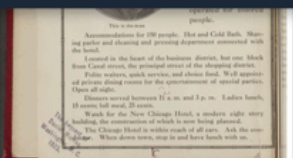
p. 4



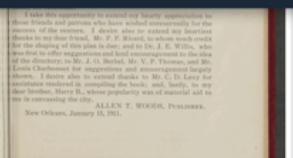
p. 5



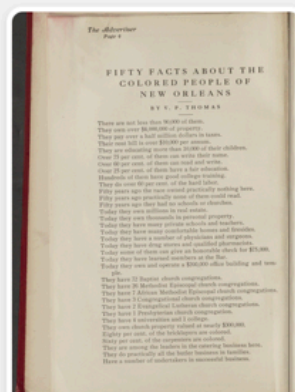
p. 6



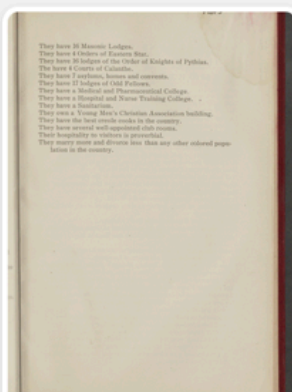
p. 7



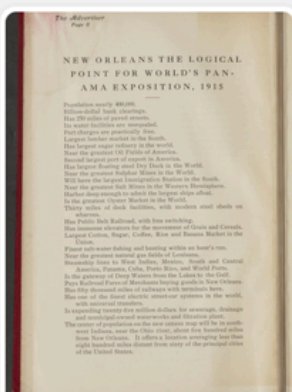
p. 8



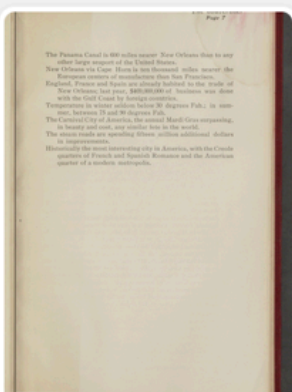
p. 9



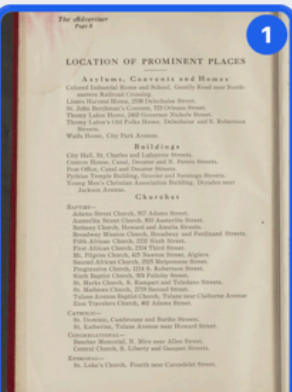
p. 10



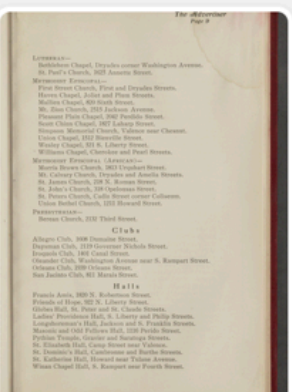
p. 11



p. 12



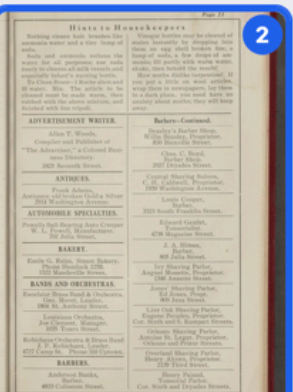
p. 13



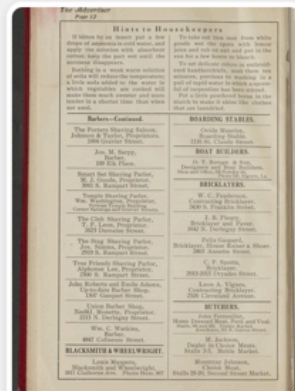
p. 14



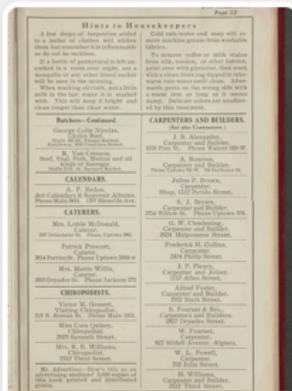
p. 15



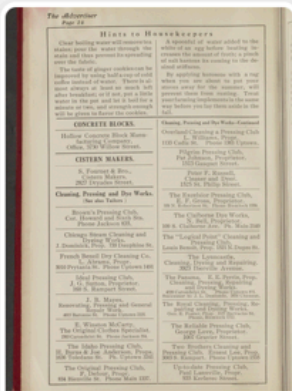
p. 16



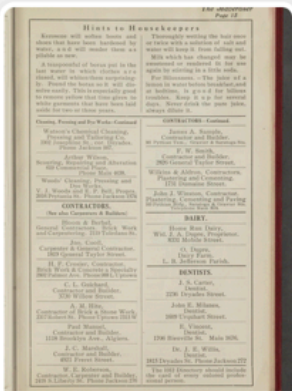
p. 17



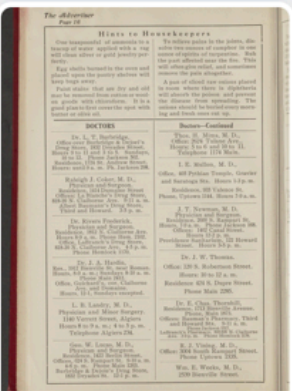
p. 18



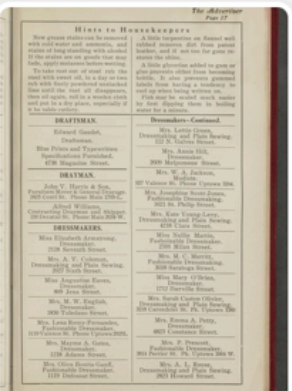
p. 19



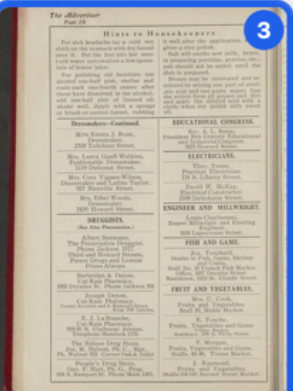
p. 20



p. 21



p. 22



p. 23



p. 24



Search page... ≥

esbrooklync1852unse_jp2.zip%2Fhearnesbrooklync1852unse_jp2%2Fhearnesbrooklync1852unse_0273.jp2
hearnesbrooklync1852unse

29 unmatched

0555_hearnesbrooklync1852unse%2Fhearnesbrooklync1852unse_jp2.zip%2Fhearnesbrooklync1852unse_jp2%2Fhearnesbrooklync1852unse_0559.jp2
hearnesbrooklync1852unse

27 unmatched

0539_hearnesbrooklync1852unse%2Fhearnesbrooklync1852unse_jp2.zip%2Fhearnesbrooklync1852unse_jp2%2Fhearnesbrooklync1852unse_0543.jp2
hearnesbrooklync1852unse

26 unmatched

0526_hearnesbrooklync1852unse%2Fhearnesbrooklync1852unse_jp2.zip%2Fhearnesbrooklync1852unse_jp2%2Fhearnesbrooklync1852unse_0530.jp2
hearnesbrooklync1852unse

25 unmatched

0545_hearnesbrooklync1852unse%2Fhearnesbrooklync1852unse_jp2.zip%2Fhearnesbrooklync1852unse_jp2%2Fhearnesbrooklync1852unse_0549.jp2
hearnesbrooklync1852unse

25 unmatched

0552_hearnesbrooklync1852unse%2Fhearnesbrooklync1852unse_jp2.zip%2Fhearnesbrooklync1852unse_jp2%2Fhearnesbrooklync1852unse_0556.jp2
hearnesbrooklync1852unse

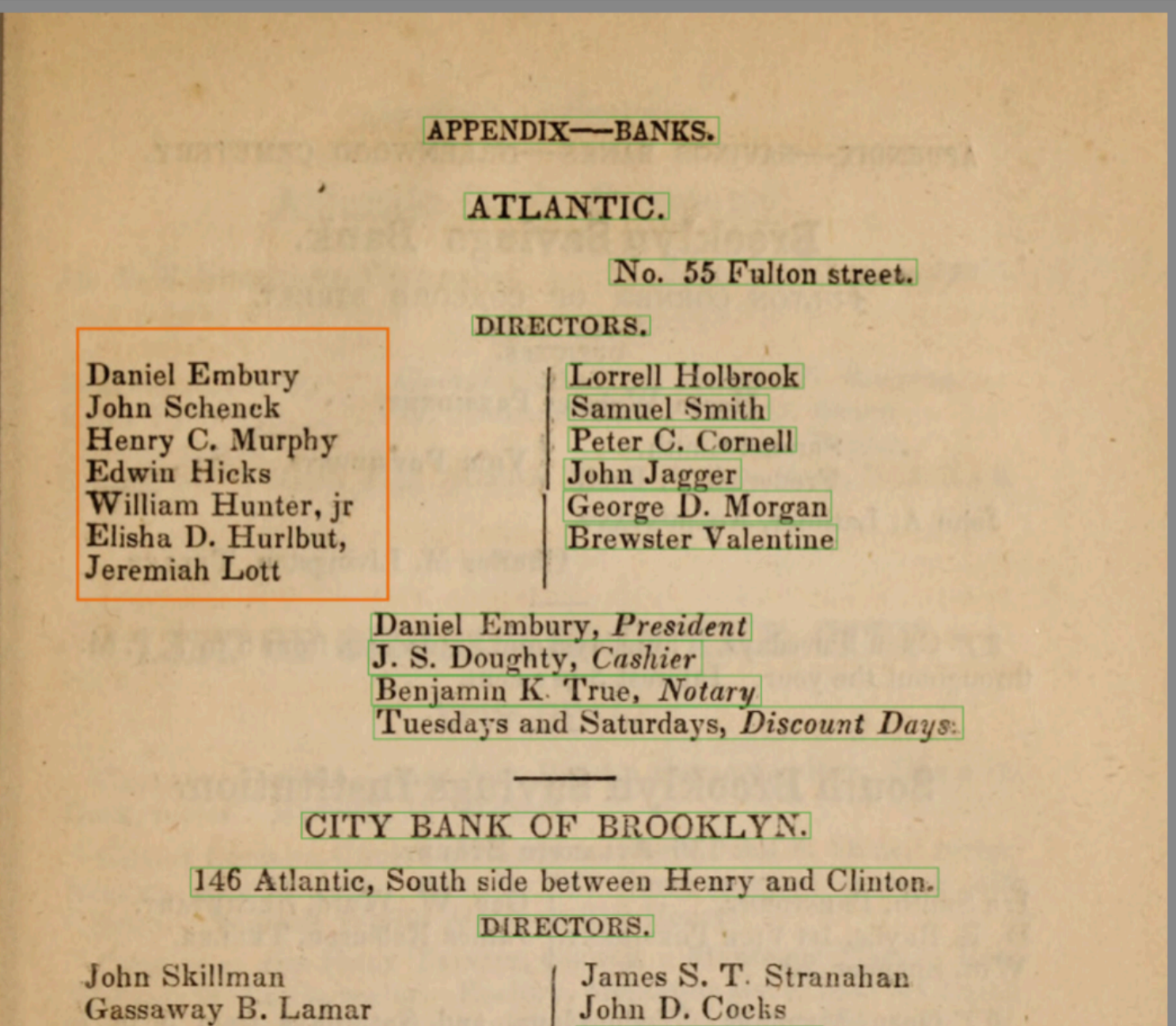
23 unmatched

0019_hearnesbrooklync1852unse%2Fhearnesbrooklync1852unse_jp2.zip%2Fhearnesbrooklync1852unse_jp2%2Fhearnesbrooklync1852unse_0019.jp2

Run Surya on box

Done — 7 Surya lines detected.

Done reviewing



Unmatched Gemini text (29)

- Daniel Embury
- John Schenck
- Henry C. Murphy
- Edwin Hicks
- William Hunter, jr
- Elisha D. Hurlbut,
- Jeremiah Lott
- John Greenwood
- Anthony P. Ostrom
- James Van Nostrand
- James S. T. Stranahan

Proposed matches — check to accept

- ☒

Daniel Embury

→ Daniel Embury

100%
- ☒

John Schenck

→ John Schenck

100%
- ☒

Henry C. Murphy

→ Henry C. Murphy

100%
- ☒

Edwin Hicks

→ Edwin Hicks

100%
- ☒

William Hunter, jr

→ William Hunter, jr

100%
- ☒

Elisha D. Hurlbut,

→ Elisha D. Hurlbut,

100%
- ☒

Jeremiah Lott

→ Jeremiah Lott

100%
- ☐

—

→ John Greenwood
- ☐

—

→ Anthony P. Ostrom
- ☐

—

→ James Van Nostrand
- ☐

—

→ James S. T. Stranahan
- ☐

—

→ John D. Cocks
- ☐

—

→ John B. Kitching

Save accepted

Discard

> python main.py output/directory_of_the_grand_and_subordinate_l_74175569 --extract-entries

Pipeline: --extract-entries
Targets: 1

[1/1] directory_of_the_grand_and_subordinate_l_74175569

uuid: None | slug: directory_of_the_grand_and_subordinate_l_74175569 | type: dir

csv: output/directory_of_the_grand_and_subordinate_l_74175569/directory_of_the_grand_and_subordinate_l_74175569.csv

imgs: output/directory_of_the_grand_and_subordinate_l_74175569/

— pipeline/extract_entries.py

\$ /Users/joshhadro/github/directory-pipeline/.venv/bin/python /Users/joshhadro/github/directory-pipeline/pipeline/extract_entries.py output/di
rectory_of_the_grand_and_subordinate_l_74175569

NER prompt: /Users/joshhadro/github/directory-pipeline/output/directory_of_the_grand_and_subordinate_l_74175569/74175569/ner_prompt.md

NOTE: no OCR files found for model 'gemini-3.1-flash-lite-preview'.

Auto-detected OCR model 'gemini-2.0-flash' in directory_of_the_grand_and_subordinate_l_74175569 – using that.

Pass --aligned-model gemini-2.0-flash to suppress this message.

Extracting entries: 1 item dir(s), model=gemini-3.1-flash-lite-preview (aligned by gemini-2.0-flash), fallback=gemini-2.5-flash, mode=text-only

NER prompt (per-item): output/directory_of_the_grand_and_subordinate_l_74175569/74175569/ner_prompt.md

74175569: 30 page(s), mode=text-only

[001/30] 0082_service:rbc:lcrbmrp:t8072:082_gemini-2.0-flash_aligned.json ok 0 entries [Key to Directory > None]

[002/30] 0084_service:rbc:lcrbmrp:t8072:084_gemini-2.0-flash_aligned.json ok 61 entries [Masonic Directory > None]

[003/30] 0086_service:rbc:lcrbmrp:t8072:086_gemini-2.0-flash_aligned.json ok 63 entries [Masonic Directory > None]

[004/30] 0088_service:rbc:lcrbmrp:t8072:088_gemini-2.0-flash_aligned.json ok 59 entries [Masonic Directory > None]

[005/30] 0090_service:rbc:lcrbmrp:t8072:090_gemini-2.0-flash_aligned.json ok 60 entries [Masonic Directory > None]

[006/30] 0092_service:rbc:lcrbmrp:t8072:092_gemini-2.0-flash_aligned.json ok 63 entries [Masonic Directory > None]

[007/30] 0094_service:rbc:lcrbmrp:t8072:094_gemini-2.0-flash_aligned.json ok 61 entries [Masonic Directory > None]

[008/30] 0096_service:rbc:lcrbmrp:t8072:096_gemini-2.0-flash_aligned.json ok 63 entries [Masonic Directory > None]

[009/30] 0098_service:rbc:lcrbmrp:t8072:098_gemini-2.0-flash_aligned.json ok 64 entries [Masonic Directory > None]

[010/30] 0100_service:rbc:lcrbmrp:t8072:100_gemini-2.0-flash_aligned.json ok 65 entries [Masonic Directory > None]

[011/30] 0102_service:rbc:lcrbmrp:t8072:102_gemini-2.0-flash_aligned.json ok 62 entries [Masonic Directory > None]

[012/30] 0104_service:rbc:lcrbmrp:t8072:104_gemini-2.0-flash_aligned.json ok 64 entries [Masonic Directory > None]

[013/30] 0106_service:rbc:lcrbmrp:t8072:106_gemini-2.0-flash_aligned.json ok 61 entries [Masonic Directory > None]

[014/30] 0108_service:rbc:lcrbmrp:t8072:108_gemini-2.0-flash_aligned.json ok 23 entries [Masonic Directory > None]

[015/30] 0110_service:rbc:lcrbmrp:t8072:110_gemini-2.0-flash_aligned.json ok 23 entries [M. W. Grand Lodge Officers > Grand Lodge]

[016/30] 0112_service:rbc:lcrbmrp:t8072:112_gemini-2.0-flash_aligned.json ok 24 entries [Lodge Officers for 1904 > Widow's Son Lodge, No. 7]

Thesis

The tradeoffs have shifted.

For digitized collections:

- Data extraction with computers used to be impossible.
- Then it was doable, but hard and expensive, and not that useful.
- Then it was doable, and relatively cheap, but useful only in narrow circumstances.

Now, I think we're on the cusp of doable, cheap, and broadly useful.

How I think about this

The baseline I'm evaluating against isn't "is this perfect" (OCR is far from perfect!). It's "is this still useful, despite known flaws." For OCR, we answered that question affirmatively decades ago. I think the answer for some collections materials in the library sector, informed by these kinds of enrichments, is getting close to yes.

Meta-prompting for item-specific extraction guidance

You are a structured data extractor for a digitized historical document. Your goal is to identify and extract discrete records from the transcribed text of "The National Directory of Morticians," a professional registry of funeral homes and directors organized by geography.

Your task

Entry schema

Each object in the "entries" array must represent a single business or practitioner. Inherit the geographic context from the headings above the entry. Use the following fields:

- state: The state name (normalized, e.g., "ALABAMA").
- city: The city name.
- county: The county name (e.g., "Henry Co.").
- city_population: *The population count listed for that city.*
- business_name: The primary name of the funeral home or mortuary.
- personnel: Names of specific directors, managers, or partners mentioned (e.g., "Bernie T. Hoff, Mgr.").

Rules

1. Extract every distinct funeral service provider listed. If a boxed advertisement and a text listing refer to the same business, merge them into a single entry containing all available details.
2. Skip page numbers, running headers, and decorative elements. Ignore generic directory filler text (e.g., "Use National Directory of Morticians for Accuracy") and "Publishers Notes."
3. Normalize headings: If a heading appears as "ARIZONA-Continued", record the state as "ARIZONA".
4. Heading transitions mid-page: When a new City/County heading appears (e.g., "BIRMINGHAM-Jefferson Co."), every entry following it belongs to that new context. The *prior_context only applies to entries appearing before the first heading change on the current page.*
5. *If a record spans a page boundary, extract the portion of the record present on the current page. If a listing says "(See Ad. next page)", include that note in the address or a notes field.*
6. *Sentinel tokens: If the source text contains [illegible] or [blank], copy those tokens verbatim. If a field is simply not present, leave it as null or an empty string.*

Return only valid JSON. No markdown code fences. No explanatory text.

Pipeline actors

What a **human** is doing in this pipeline

- Bringing curiosity to bear
- Exhibiting agency and responsibility
- Employing materials judgement and expertise
- Model selection
- Prompt review
- QA: OCR quality, alignment review, gut check etc.
- Synthesizing, understanding, and getting excited about possibilities

Pipeline actors

What an **LLM** is doing in this pipeline

- Meta-prompting for prompt-generation (OCR, NER)
- Item-specific OCR extraction
- Handwriting detection
- Item-specific data extraction

Pipeline actors

What a **plain old computer** is doing in this pipeline

- File management
- Spread detection (.e.g., microfilm with 2-up pages)
- Column detection
- Layout analysis
- Bounding box identification
- Outlier detection
- LLM OCR to bounding box alignment
- IIIF annotation and content state generation

Useful links:

Code repository: <https://github.com/hadro/directory-pipeline>

N.B. Gemini and similar API models are currently the best mix of quality and value for OCR/HTR and NER, but nearly all of these things could likely be done locally with tools like Surya/Chandra and models like Qwen 3.5, removing the paid API aspect of this.

Readings

These links substantially inspired my interest in this project:

- [Multimodal LLMs for OCR, OCR Post-Correction, and Named Entity Recognition in Historical Documents](#)
- [Gemini 3 Solves Handwriting Recognition and it's a Bitter Lesson](#)
- [The A.I. Disruption We've Been Waiting for Has Arrived](#)

AI Disclosure

Claude Code wrote:

- Most of the code for this project
- Most of the code and git repo documentation for this project

I wrote:

- Some small amount of the code for this project
- Amendments and edits of the documentation

I also wrote all of the material for this presentation — no AI was used in these slides (for better or for worse)